Standing box jumps are frequently employed in strength and conditioning programs as valuable plyometric exercises due to their low impact and effectiveness at improving explosive concentric strength and rate of force development. The relationship between vertical jumping and ground reaction force development is well researched, yet little understanding exists regarding the force production associated with standing box jumps at varying jump heights.
PURPOSE: To quantify the vertical ground reaction forces (VGRF) developed during a standing box jump at varying box heights. We hypothesized that VGRF would increase with increasing box height.
METHODS: One male (age 29 years, mass 85.2kg, height 174.5cm), completed standing box jumps at 7 different heights. Box jump heights included: 24, 30, 36, 48, 51, 54, and 57 inches. This study was limited to one participant due to the box heights employed and the ability of the athlete to complete all heights. Box jumps were completed using an arm countermovement with box positioned at a self-selected distance away from the athlete. After completing a standardized warm up jump trials at each height were initiated from a standing position on a force platform. Rest periods averaging 180 seconds were provided in between jumps to eliminate the effect of fatigue. VGRF data were collected and peak VGRF (pVGRF) in pound-force (lbf) was analyzed and then averaged for the trials at each jump height.
RESULTS: pVGRF decreased 27.4% as jump height increased 137.5%. The following values are reported: pVGRF at 24 inches = 580.5lbf; 30 inches = 569.4lbf; 36 inches = 590.0lbf; 48 inches = 527.8lbf; 51 inches = 533.0lbf; 54 inches = 519.4lbf; and 57 inches = 421.39lbf.
CONCLUSION: This data demonstrates that as box jump height increased the pVGRF decreased. This was contrary to our hypothesis and other research examining vertical jumping and pVGRF. During the jump trials it was also observed that as jump height increased arm countermovement also increased. We therefore surmise that increased upper extremity movement augmented lower extremity kinetics during the higher box heights. This theory is supported by other studies that have reported that arm motion decreased the magnitude of VGRF produced. Further research on the relationships between arm motion, pVGRF, and jump height is warranted.

1338 Board #131 May 28, 8:00 AM - 9:30 AM

Effects of Whole Body Vibration on Voluntary Torque Following Exercise Induced Muscle Damage in Recreationally Trained Females

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Voluntary torque has previously been shown to be reduced immediately after and up to 48hours following exercise induced muscle damage (EIMD). Whole-body vibration (WBV) has been shown as a modality to increase torque production and has been investigated in reducing the decrease in torque production following EIMD.
PURPOSE: The purpose of this investigation was to determine the effects of WBV on voluntary torque (VT) during maximal voluntary isometric contraction (MVIC) following EIMD in females.
METHODS: Thirty female participants (age 21± 1.9yrs, height 165.69 ± 7.3cm, mass 58.69 ± 10.95kg) volunteered to participate in 4 testing sessions. Participants were randomly assigned to either a treatment or control group. WBV or control (no WBV) was given each day. VT during a MVIC was collected before and after treatment for 3 days following EIMD via split squats. Two sets of measures were collected each day, consisting of a pre measurement followed by WBV or control and a post measurement. A 8x2 (time by group) mixed factor analysis of variance was conducted to test changes in VT.
RESULTS: No significant ($p>0.05$) interaction of group by time was found. There was no significant ($p>0.05$) main effect for group but there was a significant ($p<0.05$) main effect for time. Baseline VT was greater than post1, 24pre, 24post, 48pre, 48post, 72pre and 72post.
CONCLUSIONS: These results show a reduction in voluntary torque production following EIMD. However, the results indicate that WBV did not facilitate voluntary torque production following EIMD. Further research is needed to investigate muscle contractile properties during an MVIC to further explain the conflicting literature.

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Core Muscle Activation During Unstable Bicep Curl Using a Water-Filled Training Tube

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Stability and balance are essential components for functional movement. These components may developed by training with unstable surfaces, or unstable loads.
PURPOSE: The purpose of this study was to use a novel unstable implement ("slosh tube") and assess the degree of muscle instability created during a bicep curl.
METHODS: Ten men (age= 21± 1.6y, ht=180.0 ± 3.3cm, mass= 87.4 ± 15.0kg) and ten women (age= 19.6± 1.3y, ht=161.4 ± 12.0cm, mass= 61.2 ± 7.4kg) completed three, 30s trials of a bicep curl using an 11.4 kg tube partially filled with water. A central valve allowed three conditions of water movement: 50% open, 100% open, and a balanced, closed valve setting. Subjects completed 8-10 repetitions within each condition (1.5s per rep) using a counter-balanced design. Muscle activation was assessed on the right side, using bipolar surface electrodes placed over the belly of the bicep, deltoid, rectus abdominus and paraspinal muscles. Integrated EMG was measured for each repetition and converted to a %MVC for each muscle. Instability was determined using the coefficient of variation across repetitions. A two way repeated measures ANOVA (gender, condition, gender x condition) with post hoc t-tests was used to examine concentric contractions only.
RESULTS: Significant instability was seen in the paraspinal muscles for both genders at the 50% (CV= 26.5 ± 14.5) and 100% (CV= 27.1 ± 9.9) condition compared to stable (CV= 15.0 ± 9.1). Across all subjects the deltoid showed greatest instability at the 100% open setting (CV= 38.5 ± 23.2) compared to stable (CV= 21.0 ± 7.5). For the bicep, men showed greater overall instability (CV= 30.9 ± 17.0) compared to women (CV= 22.9 ± 8.4). There were no differences in abdominal activation in any condition. Percent MVC activated was significantly greater for women for the bicep (50= 97.7 ± 57.6, 100= 88.1 ± 55.6, stable= 87.0 ± 50.1) and paraspinal (50= 36.7 ± 22.3, 100= 47.5 ± 29.0, stable= 38.6 ± 23.9) compared to men (bicep 50= 39.5 ± 17.0, 100= 38.3 ± 18.3, stable= 31.8 ± 11.1) (paraspinal 50= 15.1 ± 10.0, 100= 16.2 ± 8.8, stable= 14.3 ± 9.6). No differences were seen for deltoid or abdominal.
CONCLUSIONS: We conclude that a bicep curl using an unstable implement utilizing water inertia causes increased paraspinal and deltoid instability in men and women, with additional instability in the bicep for men.

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Core Muscle Activation During Unstable Leg Extension Using a Water-Filled Training Tube

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Creating muscle instability may be an effective means of training for stability. Instability may be achieved using an unstable surface or load.
PURPOSE: The purpose of this study was to assess the degree of muscle instability created during a seated leg extension using a novel unstable load ("slosh tube") under three different conditions.
METHODS: Eight men (age= 19.9± 1.6y, ht=178.8 ± 8.9cm, mass= 103.3 ± 48.5kg, Leg Ext 1RM= 136.5 ± 15.9kg) and eight women (age= 20.1 ± 1.1y, ht=164.9 ± 9.8cm, mass= 62.9 ± 12.2kg, Leg Ext 1RM= 91.5 ± 15.5kg) completed three, 30s trials of leg extension using an 11.4 kg tube partially filled with water. A central valve allowed three conditions of water movement: 50% open, 100% open, and a stable, balanced valve setting. The valve settings allowed for different water flow dynamics, creating movement during the lift. Subjects completed 8-

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